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Electro-Magnetic Field (force caused by potential)

- static: lovslized source charge at rest, only spacial dependency, no time dependency
- stationary: source charce at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistc: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movemnt
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c}\left(\frac{r}{|r|} \times \mathcal{E}\right) \overset{\text{Far Field}}{=} \frac{1}{2}\frac{1}{2\pi}\mu_0 \frac{k^2 e^{i(kr-\omega t)}}{r}\frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_{0_{dipol}} \times \frac{r}{|r|}\right) \overset{\text{Near Field}}{\approx} \frac{1}{2}\frac{1}{2\pi}\mu_0 q \frac{v \times r}{|r|^2} \tag{1}$$

Magnetic Field in Matter

Exitation, Displacement

$$\mathcal{H} = ... \tag{2}$$

Electric Field in Vacuum

$$\mathcal{E} \overset{\text{conservativ}}{=} -\nabla U_e \overset{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \overset{\text{fast moving charge}}{=} \frac{1}{2}\frac{1}{2\pi}\frac{1}{\varepsilon_0}\frac{1}{r}\frac{1}{\left(1 - \frac{r}{|r|}\cdot\frac{v}{c}\right)^3} q \left[\underbrace{\frac{\left(1 - \frac{v^2}{c^2}\right)\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right.\right.}{}}_{\text{accelerate}} \right. \tag{3}$$

Electric Field in Matter

Exitation, Displacement (field change)
dielectricum = non-conducting material
electricum = conductor medium

$$\mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \quad \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \varepsilon_0 \langle \mathcal{E}_m \rangle = \alpha n \varepsilon_0 \tag{4}$$

Abbreviations

$$\mathcal{E} = \text{Electric Field} \qquad \mathcal{B} = \text{Magnetic Fiels} \qquad \mathcal{D} = \text{Displacement density of dielectricum}$$

$$\mathcal{P} = \text{Polarisation} \qquad \mathcal{H} = \text{magn. exitation, magn. field strength ("magn. displacement")}$$

$$\mathcal{M} = \text{magnetisation density} \qquad n = \text{spacial molecule density in volume}$$

$$\alpha = 4\pi r^3 = constant \qquad \chi = \text{elec. susceptibility of dieelectricum} \qquad \kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$$

$$\langle p_m \rangle = \text{spacial average of magn. dipol moment} \qquad \langle \mathcal{E}_m \rangle = \text{spacial average of electric field}$$

$$\bar{\mathcal{E}}_m = \text{time average of electric field} \qquad \mu = \text{magn. permeability} \qquad \mu_0 = \text{magn. constant ...}$$

$$\varepsilon = \text{dielectricity constant} \qquad \varepsilon_0 = \text{...}$$